Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/16/2007
PI Name:	Revankar, Shripad T Ph.D.		
Project Title:	Study of Co-Current and Counter-Current Gas-Liquid Two-Phase Flow Through Packed Bed in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907	**Congressional District:**	4
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
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	The present research addresses the most important topic of HEDS-specific microgravity fluid physics research identified by NASA‘s one of the strategic enterprises, OBPR Enterprise. The main goal of the project is to obtain new experimental data and development of models on the co-current and counter-current gas-liquid two-phase flow through a packed bed in microgravity and characterize the flow regime transition, pressure drop, void and interfacial area distribution, and liquid hold up. An experimental and analytical modeling program for the study of co-current air-water two-phase down flow in a packed column was developed. This report presents the progress made in the last year, 2004-2005. Over the last year we have refined the instrumentation for local measurement in trickle bed reactors and developed more accurate calibration methods which are presented. Experiments were conducted in a cylindrical packed bed. The bed is a cylindrical pipe of diameter 69 mm and height of 1.52m. The bed particles were glass spheres of diameter 6 mm. The bed porosity was 0.377. A two point conductivity probe for the measurement of void fraction in the bed at pore level, a film thickness probe for the measurement of film thickness and drying characteristics of the packing, and a bed wise impedance meter for bed wise void measurement were used. Refined designs for flush film thickness probes, conductivity probes, and impedance meters were developed. The modeling effort included development of drift flux model for the packed bed and model for packed bed two-phase flow in microgravity. The drift flux model was developed by averaging the properties of the two-phase (gas-liquid) flow characteristics over bed cross section. The drift velocity and the distribution parameters were obtained using experimental data on phase flux and the void fraction. The drift flux parameters were obtained for trickle, pulse, and bubbly flow regimes. The distribution parameter was found to vary between 1.1 and 1.3. The drift velocity showed negative value for the trickle floe regime. A survey of literature shows only one set of study from Glenn Research Center NASA is available on two-phase flow in packed bed under microgavity conditions. Models in the literature for pressure drop and liquid hold up were evaluated. The models were then modified so that they can be applied to the microgravity condition. The existing correlations for the liquid hold up at the gravity conditions were modified by examining the microgravity data and a new correlation for liquid hold up was obtained for microgravity condition. The pressure drop and liquid hold up for paced two-phase flow in microgravity were predicted for pulse flow and were compared with the NASA data.
Task Description:	The specific objectives of the proposed research are to: 1. Develop experiments for the study of the gas liquid two-phase flow through the packed bed with three different flow combinations: co-current down flow, co-current upflow and counter current flow. 2. Develop pore scale and bed scale two-phase instrumentation for measurement of flow regime transition, void distribution and gas-liquid interfacial area concentration in the packed bed. 3. Obtain database on flow regime transition, pressure drop, void distribution, interfacial area concentration and liquid hold up as a function of bed characteristics such as bed particle size, porosity, and liquid properties such as viscosity and surface tension. 4. Develop mathematical model for flow regime transition, void fraction distribution and interfacial area concentration for co-current gas-liquid flow through the porous bed in gravity and micro gravity conditions. 5. Develop mathematical model for the flooding phenomena in counter-current gas-liquid flow through the porous bed in gravity and micro gravity conditions. 6. Train graduate and undergraduate students in instrumentation and measurement methods with particular emphasis on two-phase flow through packed beds. Packed bed reactors are probably the oldest and most widely used catalytic chemical reactors. Packed beds are used for both single phase and multi-phase catalytic processes. Gas-liquid two-phase flow through a fixed bed of particles occurs in many unit operations of interest to the designers of space-based as well as terrestrial equipment. Examples include separation columns, gas-liquid reactors, humidification, drying and gas absorption operations and extraction and leaching of minerals. Trickle-bed reactors, in which a gas and a liquid flow cocurrently downward over a randomly packed stationary bed, are widely used in variety of gas-liquid -solid catalytic reactions including petroleum and petrochemical industries. Potential space-based applications include: fuel cells, bioregenerative food production, waste treatment, watering and nutrient transport in plant root systems, waste recovery, and extraction from planetary materials. In general, gravitational, capillary, and viscous forces drive flow in porous media. Gravity causes phase separation and migration in the direction of the gravitational field. In microgravity, capillary and viscous forces play fundamental role in controlling phase distribution and hence, multiphase flow and transport in packed beds. Flow regime, pressure drop, flow stability and dryout, therefore, are also expected to be markedly different in microgravity. Though recently various studies have been conducted for two-phase flow through packed bed at earth gravity condition to understand the mechanisms governing the flow regime transition, pressure drop, void and interfacial area distribution, there is no published work available on this topic at microgravity condition. Also the pore level study of the two-phase flow through packed beds has not been investigated in detail even at earth gravity condition. The bed characterization is generally done through the macro scale parameters such as total bed pressure drop, liquid hold up and total void. It should be noted that the macro scale parameters are rooted in the micro scale, where two-phases compete for the void space. In view of this an experimental and analytical study is carried out where systematic experiments are conducted for the gas-liquid two-phase flow in a packed bed in earth gravity and microgravity conditions. Data on key bed flow parameters are obtained both at micro scale and macro scale. Modeling efforts take account of the two-phase hydrodynamics in micro scale (pore) level. It is expected that the results of this modeling effort will eventually lead to better understanding of the bed characteristics at the microgravity conditions and hence support safe and efficient packed bed designs for space applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The present research addresses the most important topic of HEDS-specific microgravity fluid physics research identified by NASA‘s one of the strategic enterprises, OBPR Enterprise. The main goal of the project was to obtain new experimental data and development of models on the co-current and counter-current gas-liquid two-phase flow through a packed bed in microgravity and characterize the flow regime transition, pressure drop, void and interfacial area distribution, and liquid hold up. During the period of the project, instrumentation for local measurement in trickle bed reactors were developed. Flush mounted film thickness was developed which was mounted on the 6 mm sphere (packed bed particle. This probe can measure film thickness from 50 micrometer to 1 mm. A parallel wire film thickness probe was developed that could measure the film thickness from 1 mm to 3 mm on the spherical particles. A two point conductivity probe was developed for measurement of local interfacial area concentration in packed bed. The probes were implemented in 6mm packed bed. (4) A capacitance meter was developed which can be mounted external to the packed bed. This meter can measure pipe cross sectional averaged void fraction, detect flow regimes in packed bed, and measure pulse frequencies in two-phase flow in packed bed. In this report a survey of additional literature for microgravity two-phase flow and terrestrial trickle bed two-phase flow

	is presented. Design and parameters of present experiment loop, experiment procedure, and data processing are introduced. Experiments were conducted in a cylindrical packed bed. Traditional global experiment data for flow regime map and pressure drop in trickle bed is presented along side local film thickness data. The modeling of two-phase flow in trickle bed under microgravity is presented. Recent studies on modeling of interfacial area in packed bed are also presented. A wavy annular flow model was developed by introducing the shape of waves in a thin liquid film to predict the local interfacial area for the trickling flow in a packed-bed reactor. The local interfacial area predicted by the wavy annular flow model in the trickling down flow is moderately higher when compared with empirical correlations developed by former researchers. The result shows that the interfacial area increases as the flow regime changes from low to high interaction flow regime.
Bibliography Type:	Description: (Last Updated: 12/17/2007)
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